Huawei Access Node WFD-ACC-18 Optical Transport Failure SOP

Node: WFD-ACC-18 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_2454

Description

The Huawei Access Node WFD-ACC-18 at the Whitefield POP is equipped with an Optical Transport Network (OTN) interface, which employs SFP/XFP transceivers to enable high-speed data transmission over optical fibers. In the event of an Optical Transport Failure, the Role: Access node may experience issues with Rx/Tx power loss or degradation, resulting in Alarm Code ALARM_2454. This minor severity alarm signifies a potential loss of signal quality or integrity. To diagnose and resolve this issue, it is essential to understand the underlying protocol mechanics of OTN, which relies on the Optical Transport Unit (OTU) and Optical Channel Unit (OCU) to manage data transmission. The Huawei equipment's SFP/XFP transceivers are responsible for converting electrical signals to optical signals and vice versa. Fiber cuts or SFP/XFP degradation can cause Rx/Tx power loss, leading to the alarm condition. To address this failure domain, it is crucial to perform a thorough analysis of the OTN interface, including the SFP/XFP transceivers, to identify the root cause of the issue.

Pre-requisite Checks

- Verify that the Whitefield POP is accessible and that all necessary test equipment is available.
- Check the redundant link status to ensure that it is functioning correctly.
- Dispatch remote hands to the Whitefield POP to assist with the troubleshooting process.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display evn otu-info, Expected Output: OTU information for the affected interface, Description: This command displays the OTU information for the affected interface, including the OTU level, OTU rate, and OTU mode. It helps to verify the OTU configuration and identify potential issues.

Command: display evn odu-info, Expected Output: ODU information for the affected interface, Description: This command displays the ODU information for the affected interface, including the ODU level, ODU rate, and ODU mode. It helps to verify the ODU configuration and identify potential issues.
```

```
Command: display int optical transceiver, Expected Output: Optical transceiver information for the affected interface, Description: This command displays the optical transceiver information for the affected interface, including the SFP/XFP type, wavelength, and optical power. It helps to verify the optical transceiver configuration and identify potential issues.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceiver, Command: replace evn otu with , Description: Replace the SFP/XFP transceiver with a new one to resolve the issue. Ensure that the new transceiver is compatible with the Huawei equipment and the OTN interface.

Step 2: Step: Check and repair fiber cuts, Command: display evn fiber-info, Description: Check the fiber information to identify any cuts or damage. If a cut is detected, repair the fiber to resolve the issue.

Step 3: Step: Adjust the Rx/Tx power level, Command: set evn otu rx-power tx-power , Description: Adjust the Rx/Tx power level to resolve the issue. Ensure that the new power levels are compatible with the Huawei equipment and the OTN interface.

Post-Check Verification

```
Command: display evn otu-info, Expected Output: OTU information for the affected interface (resolves to normal), Description: Verify that the OTU configuration and information have been restored to normal.

Command: display evn odu-info, Expected Output: ODU information for the affected interface (resolves to normal), Description: Verify that the ODU configuration and information have been restored to normal.

Command: display int optical transceiver, Expected Output: Optical transceiver information for the affected interface (resolves to normal), Description: Verify that the optical transceiver configuration and information have been restored to normal.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: replace evn otu with , Description: Restore the original SFP/XFP transceiver to its previous state in case the new transceiver did not resolve the issue.

- Step: Restore the original Rx/Tx power levels, Command: set evn otu rx-power tx-power , Description: Restore the original Rx/Tx power levels to their previous state in case the new power levels did not resolve the issue.